

3 Differentiation Rules



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3.1

Derivatives of Polynomials and Exponential Functions

Derivatives of Polynomials and Exponential Functions (1 of 1)

In this section we learn how to differentiate constant functions, power functions, polynomials, and exponential functions.

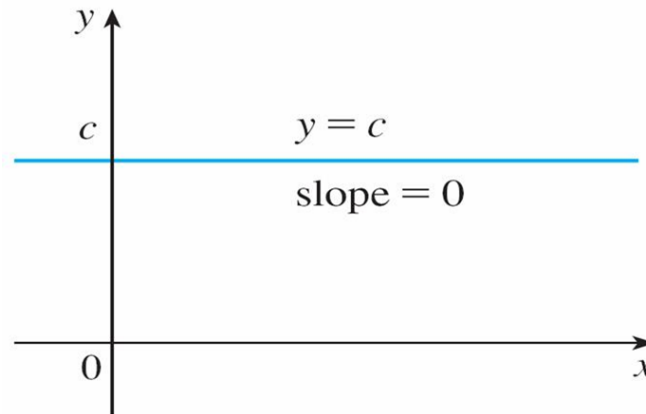


Constant Functions

Constant Functions (1 of 2)

Let's start with the simplest of all functions, the constant function $f(x) = c$.

The graph of this function is the horizontal line $y = c$, which has slope 0, so we must have $f'(x) = 0$. (See Figure 1.)



The graph of $f(x) = c$ is the line $y = c$,
so $f'(x) = 0$.

Figure 1

Constant Functions (2 of 2)

A formal proof, from the definition of a derivative, is also easy:

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} = \lim_{h \rightarrow 0} \frac{c - c}{h} = \lim_{h \rightarrow 0} 0 = 0$$

In Leibniz notation, we write this rule as follows.

Derivative of a Constant Function

$$\frac{d}{dx}(c) = 0$$



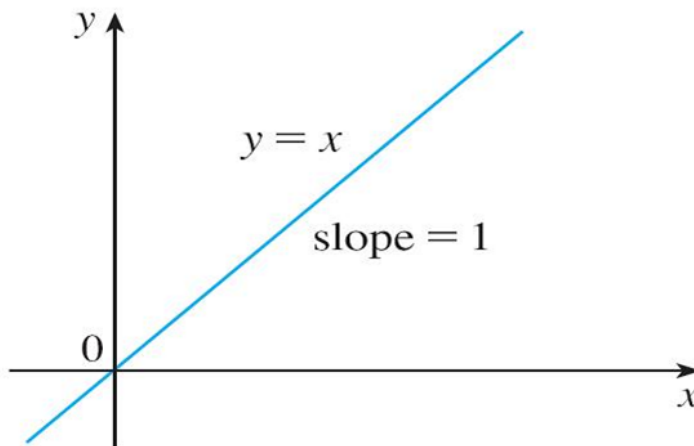
Power Functions

Power Functions (1 of 5)

We next look at the functions

$$f(x) = x^n, \text{ where } n \text{ is a positive integer.}$$

If $n = 1$, the graph of $f(x) = x$ is the line $y = x$, which has slope 1. (See Figure 2.)



The graph of $f(x) = x$ is the line $y = x$, so $f'(x) = 1$.

Figure 2

Power Functions (2 of 5)

So

$$1 \quad \frac{d}{dx}(x) = 1$$

(You can also verify Equation 1 from the definition of a derivative.) We have already investigated the cases $n = 2$ and $n = 3$. We found that

$$2 \quad \frac{d}{dx}(x^2) = 2x \quad \frac{d}{dx}(x^3) = 3x^2$$

Power Functions (3 of 5)

For $n = 4$ we find the derivative of $f(x) = x^4$ as follows:

$$\begin{aligned} f'(x) &= \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} \\ &= \lim_{h \rightarrow 0} \frac{(x+h)^4 - x^4}{h} \\ &= \lim_{h \rightarrow 0} \frac{x^4 + 4x^3h + 6x^2h^2 + 4xh^3 + h^4 - x^4}{h} \\ &= \lim_{h \rightarrow 0} \frac{4x^3h + 6x^2h^2 + 4xh^3 + h^4}{h} \\ &= \lim_{h \rightarrow 0} (4x^3 + 6x^2h + 4xh^2 + h^3) \\ &= 4x^3 \end{aligned}$$

Power Functions (4 of 5)

Thus

$$3 \quad \frac{d}{dx}(x^4) = 4x^3$$

Comparing the equations in (1), (2), and (3), we see a pattern emerging. It seems to be a reasonable guess that, when n is a positive integer, $(d/dx)(x^n) = nx^{n-1}$. This turns out to be true.

The Power Rule If n is a positive integer, then

$$\frac{d}{dx}(x^n) = nx^{n-1}$$

Example 1

(a) If $f(x) = x^6$, then $f'(x) = 6x^5$.

(b) If $y = x^{1000}$, then $y' = 1000x^{999}$.

(c) If $y = t^4$, then $\frac{dy}{dt} = 4t^3$.

(d) $\frac{d}{dr}(r^3) = 3r^2$

Power Functions (5 of 5)

The Power Rule (General Version) If n is any real number, then

$$\frac{d}{dx}(x^n) = nx^{n-1}$$

The Power Rule enables us to find tangent lines without having to resort to the definition of a derivative. It also enables us to find *normal lines*.

The **normal line** to a curve C at a point P is the line through P that is perpendicular to the tangent line at P .



New Derivatives from Old

New Derivatives from Old (1 of 3)

When new functions are formed from old functions by addition, subtraction, or multiplication by a constant, their derivatives can be calculated in terms of derivatives of the old functions.

In particular, the following formula says that *the derivative of a constant times a function is the constant times the derivative of the function*.

The Constant Multiple Rule If c is a constant and f is a differentiable function, then

$$\frac{d}{dx} [cf(x)] = c \frac{d}{dx} f(x)$$

Example 4

$$\begin{aligned}\text{(a)} \quad \frac{d}{dx}(3x^4) &= 3 \frac{d}{dx}(x^4) \\ &= 3(4x^3) \\ &= 12x^3\end{aligned}$$

$$\begin{aligned}\text{(b)} \quad \frac{d}{dx}(-x) &= \frac{d}{dx}[(-1)x] \\ &= (-1) \frac{d}{dx}(x) \\ &= -1(1) \\ &= -1\end{aligned}$$

New Derivatives from Old (2 of 3)

The next rule tells us that *the derivative of a sum (or difference) of functions is the sum (or difference) of the derivatives.*

The Sum and Difference Rules If f and g are both differentiable, then

$$\frac{d}{dx} [f(x) + g(x)] = \frac{d}{dx} f(x) + \frac{d}{dx} g(x)$$

$$\frac{d}{dx} [f(x) - g(x)] = \frac{d}{dx} f(x) - \frac{d}{dx} g(x)$$

The Sum Rule can be extended to the sum of any number of functions. For instance, using this theorem twice, we get

$$(f + g + h)' = [(f + g) + h]' = (f + g)' + h' = f' + g' + h'$$

New Derivatives from Old (3 of 3)

The Constant Multiple Rule, the Sum Rule, and the Difference Rule can be combined with the Power Rule to differentiate any polynomial.



Exponential Functions

Exponential Functions (1 of 7)

Let's try to compute the derivative of the exponential function using the definition of a derivative:

$$f(x) = b^x$$

$$\begin{aligned} f'(x) &= \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} = \lim_{h \rightarrow 0} \frac{b^{x+h} - b^x}{h} \\ &= \lim_{h \rightarrow 0} \frac{b^x b^h - b^x}{h} = \lim_{h \rightarrow 0} \frac{b^x (b^h - 1)}{h} \end{aligned}$$

The factor b^x doesn't depend on h , so we can take it in front of the limit:

$$f'(x) = b^x \lim_{h \rightarrow 0} \frac{b^h - 1}{h}$$

Exponential Functions (2 of 7)

Notice that the limit is the value of the derivative of f at 0, that is,

$$\lim_{h \rightarrow 0} \frac{b^h - 1}{h} = f'(0)$$

Therefore we have shown that if the exponential function $f(x) = b^x$ is differentiable at 0, then it is differentiable everywhere and

$$4 \quad f'(x) = f'(0) b^x$$

This equation says that *the rate of change of any exponential function is proportional to the function itself*. (The slope is proportional to the height.)

Exponential Functions (3 of 7)

Numerical evidence for the existence of $f'(0)$ is given in the table shown below for the cases $b = 2$ and $b = 3$. (Values are stated correct to four decimal places.) It appears that the limits exist and

$$\text{for } b = 2, \quad f'(0) = \lim_{h \rightarrow 0} \frac{2^h - 1}{h} \approx 0.693$$

$$\text{for } b = 3, \quad f'(0) = \lim_{h \rightarrow 0} \frac{3^h - 1}{h} \approx 1.099$$

h	$\frac{2^h - 1}{h}$	$\frac{3^h - 1}{h}$
0.1	0.71773	1.16123
0.01	0.69556	1.10467
0.001	0.69339	1.09922
0.0001	0.69317	1.09867
0.00001	0.69315	1.09862

Exponential Functions (4 of 7)

Thus, from Equation 4, we have

$$5 \quad \frac{d}{dx}(2^x) \approx (0.693)2^x \quad \frac{d}{dx}(3^x) \approx (1.099)3^x$$

Of all possible choices for the base b in Equation 4, the simplest differentiation formula occurs when $f'(0) = 1$.

Exponential Functions (5 of 7)

In view of the estimates of $f'(0)$ for $b = 2$ and $b = 3$, it seems reasonable that there is a number b between 2 and 3 for which $f'(0) = 1$.

It is traditional to denote this value by the letter e . Thus we have the following definition.

Definition of the Number e

e is the number such that
$$\lim_{h \rightarrow 0} \frac{e^h - 1}{h} = 1$$

Exponential Functions (6 of 7)

Geometrically, this means that of all the possible exponential functions $y = b^x$, the function $f(x) = e^x$ is the one whose tangent line at $(0, 1)$ has a slope $f'(0)$ that is exactly 1. (See Figures 6 and 7.)

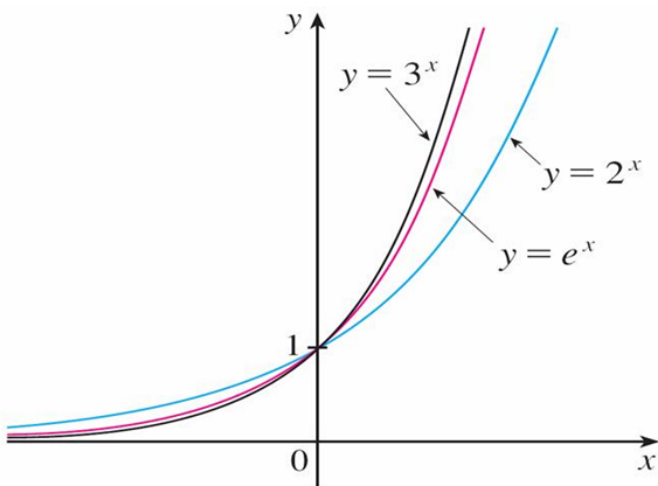


Figure 6

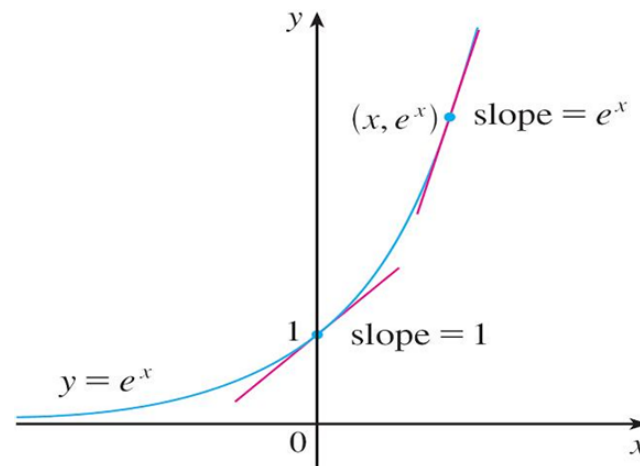


Figure 7

Exponential Functions (7 of 7)

If we put $b = e$ and, therefore, $f'(0) = 1$ in Equation 4, it becomes the following important differentiation formula.

Derivative of the Natural Exponential Function

$$\frac{d}{dx}(e^x) = e^x$$

Thus the exponential function $f(x) = e^x$ has the property that it is its own derivative. The geometrical significance of this fact is that the slope of a tangent line to the curve $y = e^x$ at a point (x, e^x) is equal to the y -coordinate of the point (see Figure 7).

Example 8

If $f(x) = e^x - x$, find f' and f'' . Compare the graphs of f and f' .

Solution:

Using the Difference Rule, we have

$$\begin{aligned} f'(x) &= \frac{d}{dx}(e^x - x) \\ &= \frac{d}{dx}(e^x) - \frac{d}{dx}(x) \\ &= e^x - 1 \end{aligned}$$

Example 8 – Solution (1 of 2)

We defined the second derivative as the derivative of f' , so

$$\begin{aligned} f''(x) &= \frac{d}{dx}(e^x - 1) \\ &= \frac{d}{dx}(e^x) - \frac{d}{dx}(1) \\ &= e^x \end{aligned}$$

Example 8 – Solution (2 of 2)

The function f and its derivative f' are graphed in Figure 8.

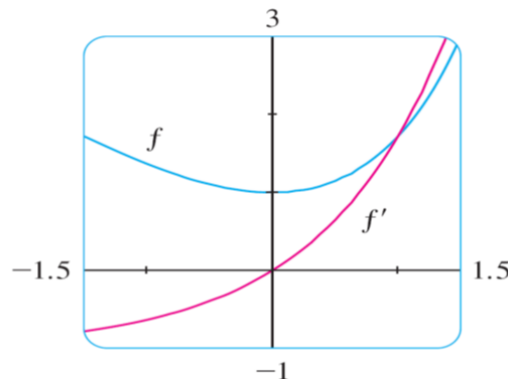


Figure 8

Notice that f has a horizontal tangent when $x = 0$; this corresponds to the fact that $f'(0) = 0$. Notice also that, for $x > 0$, $f'(x)$ is positive and f is increasing. When $x < 0$, $f'(x)$ is negative and f is decreasing.



3.2

The Product and Quotient Rules



The Product Rule

The Product Rule (1 of 6)

By analogy with the Sum and Difference Rules, we might be tempted to guess, as Leibniz did three centuries ago, that the derivative of a product is the product of the derivatives.

We can see, however, that this guess is wrong by looking at a particular example.

Let $f(x) = x$ and $g(x) = x^2$. Then the Power Rule gives $f'(x) = 1$ and $g'(x) = 2x$.

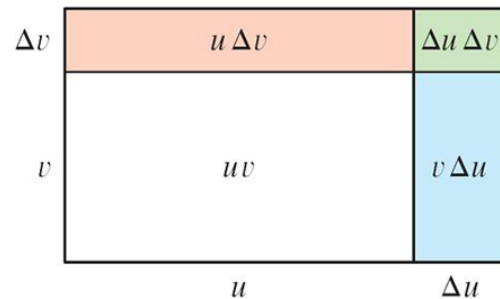
But $(fg)(x) = x^3$, so $(fg)'(x) = 3x^2$. Thus $(fg)' \neq f'g'$.

The Product Rule (2 of 6)

The correct formula was discovered by Leibniz (soon after his false start) and is called the Product Rule.

Before stating the Product Rule, let's see how we might discover it.

We start by assuming that $u = f(x)$ and $v = g(x)$ are both positive differentiable functions. Then we can interpret the product uv as an area of a rectangle (see Figure 1).



The geometry of the Product Rule

Figure 1

The Product Rule (3 of 6)

If x changes by an amount Δx , then the corresponding changes in u and v are

$$\Delta u = f(x + \Delta x) - f(x) \quad \Delta v = g(x + \Delta x) - g(x)$$

and the new value of the product, $(u + \Delta u)(v + \Delta v)$, can be interpreted as the area of the large rectangle in Figure 1 (provided that Δu and Δv happen to be positive).

The change in the area of the rectangle is

$$\begin{aligned} \textcolor{red}{1} \quad \Delta(uv) &= (u + \Delta u)(v + \Delta v) - uv = u\Delta v + v\Delta u + \Delta u\Delta v \\ &= \text{the sum of the three shaded areas} \end{aligned}$$

The Product Rule (4 of 6)

If we divide by Δx , we get

$$\frac{\Delta(uv)}{\Delta x} = u \frac{\Delta v}{\Delta x} + v \frac{\Delta u}{\Delta x} + \Delta u \frac{\Delta v}{\Delta x}$$

If we now let $\Delta x \rightarrow 0$, we get the derivative of uv :

$$\begin{aligned} \frac{d}{dx}(uv) &= \lim_{\Delta x \rightarrow 0} \frac{\Delta(vu)}{\Delta x} = \lim_{\Delta x \rightarrow 0} \left(u \frac{\Delta v}{\Delta x} + v \frac{\Delta u}{\Delta x} + \Delta u \frac{\Delta v}{\Delta x} \right) \\ &= u \lim_{\Delta x \rightarrow 0} \frac{\Delta v}{\Delta x} + v \lim_{\Delta x \rightarrow 0} \frac{\Delta u}{\Delta x} + \left(\lim_{\Delta x \rightarrow 0} \Delta u \right) \left(\lim_{\Delta x \rightarrow 0} \frac{\Delta v}{\Delta x} \right) \\ &= u \frac{dv}{dx} + v \frac{du}{dx} + 0 \cdot \frac{dv}{dx} \end{aligned}$$

The Product Rule (5 of 6)

$$2 \quad \frac{d}{dx}(uv) = u \frac{dv}{dx} + v \frac{du}{dx}$$

(Notice that $\Delta u \rightarrow 0$ as $\Delta x \rightarrow 0$ since f is differentiable and therefore continuous.)

Although we started by assuming (for the geometric interpretation) that all the quantities are positive, we notice that Equation 1 is always true. (The algebra is valid whether u , v , Δu , and Δv are positive or negative.)

The Product Rule (6 of 6)

So we have proved Equation 2, known as the Product Rule, for all differentiable functions u and v .

The Product Rule If f and g are both differentiable, then

$$\frac{d}{dx} [f(x) g(x)] = f(x) \frac{d}{dx} [g(x)] + g(x) \frac{d}{dx} [f(x)]$$

In words, the Product Rule says that *the derivative of a product of two functions is the first function times the derivative of the second function plus the second function times the derivative of the first function.*

Example 1

(a) If $f(x) = xe^x$, find $f'(x)$.

(b) Find the n th derivative, $f^{(n)}(x)$.

Solution:

(a) By the Product Rule, we have

$$\begin{aligned} f'(x) &= \frac{d}{dx}(xe^x) \\ &= x \frac{d}{dx}(e^x) + e^x \frac{d}{dx}(x) \\ &= xe^x + e^x \cdot 1 = (x+1)e^x \end{aligned}$$

Example 1 – Solution (1 of 2)

(b) Using the Product Rule a second time, we get

$$\begin{aligned} f''(x) &= \frac{d}{dx}[(x+1)e^x] \\ &= (x+1)\frac{d}{dx}(e^x) + e^x\frac{d}{dx}(x+1) \\ &= (x+1)e^x + e^x \cdot 1 \\ &= (x+2)e^x \end{aligned}$$

Example 1 – Solution (2 of 2)

Further applications of the Product Rule give

$$f'''(x) = (x + 3)e^x \quad f^{(4)}(x) = (x + 4)e^x$$

In fact, each successive differentiation adds another e^x , so term

$$f^{(n)}(x) = (x + n)e^x$$



The Quotient Rule

The Quotient Rule (1 of 4)

We find a rule for differentiating the quotient of two differentiable functions $u = f(x)$ and $v = g(x)$ in much the same way that we found the Product Rule.

If x , u , and v change by amounts Δx , Δu , and Δv , then the corresponding change in the quotient $\frac{u}{v}$ is

$$\begin{aligned}\Delta \left(\frac{u}{v} \right) &= \frac{u + \Delta u}{v + \Delta v} - \frac{u}{v} = \frac{(u + \Delta u)v - u(v + \Delta v)}{v(v + \Delta v)} \\ &= \frac{v\Delta u - u\Delta v}{v(v + \Delta v)}\end{aligned}$$

The Quotient Rule (2 of 4)

so

$$\frac{d}{dx} \left(\frac{u}{v} \right) = \lim_{\Delta x \rightarrow 0} \frac{\Delta \left(\frac{u}{v} \right)}{\Delta x} = \lim_{\Delta x \rightarrow 0} \frac{v \frac{\Delta u}{\Delta x} - u \frac{\Delta v}{\Delta x}}{v(v + \Delta v)}$$

As $\Delta x \rightarrow 0$, $\Delta v \rightarrow 0$ also, because $v = g(x)$ is differentiable and therefore continuous.

Thus, using the Limit Laws, we get

$$\frac{d}{dx} \left(\frac{u}{v} \right) = \frac{v \lim_{\Delta x \rightarrow 0} \frac{\Delta u}{\Delta x} - u \lim_{\Delta x \rightarrow 0} \frac{\Delta v}{\Delta x}}{v \lim_{\Delta x \rightarrow 0} (v + \Delta v)} = \frac{v \frac{du}{dx} - u \frac{dv}{dx}}{v^2}$$

The Quotient Rule (3 of 4)

The Quotient Rule If f and g are differentiable, then

$$\frac{d}{dx} \left[\frac{f(x)}{g(x)} \right] = \frac{g(x) \frac{d}{dx} [f(x)] - f(x) \frac{d}{dx} [g(x)]}{[g(x)]^2}$$

In words, the Quotient Rule says that the *derivative of a quotient is the denominator times the derivative of the numerator minus the numerator times the derivative of the denominator, all divided by the square of the denominator.*

Example 4

Let $y = \frac{x^2 + x - 2}{x^3 + 6}$. Then

$$\begin{aligned}y' &= \frac{(x^3 + 6) \frac{d}{dx}(x^2 + x - 2) - (x^2 + x - 2) \frac{d}{dx}(x^3 + 6)}{(x^3 + 6)^2} \\&= \frac{(x^3 + 6)(2x + 1) - (x^2 + x - 2)(3x^2)}{(x^3 + 6)^2} \\&= \frac{(2x^4 + x^3 + 12x + 6) - (3x^4 + 3x^3 - 6x^2)}{(x^3 + 6)^2} \\&= \frac{-x^4 - 2x^3 + 6x^2 + 12x + 6}{(x^3 + 6)^2}\end{aligned}$$

The Quotient Rule (4 of 4)

Table of Differentiation Formulas

$$\frac{d}{dx}(c) = 0$$

$$\frac{d}{dx}(x^n) = nx^{n-1}$$

$$\frac{d}{dx}(e^x) = e^x$$

$$(cf)' = cf'$$

$$(f + g)' = f' + g'$$

$$(f - g)' = f' - g'$$

$$(fg)' = fg' + gf'$$

$$\left(\frac{f}{g}\right)' = \frac{gf' - fg'}{g^2}$$



3.3

Derivatives of Trigonometric Functions

Derivatives of Trigonometric Functions

In particular, it is important to remember that when we talk about the function f defined for all real numbers x by

$$f(x) = \sin x$$

it is understood that $\sin x$ means the sine of the angle whose *radian* measure is x . A similar convention holds for the other trigonometric functions $\cos$, $\tan$, $\csc$, $\sec$, and $\cot$.

All of the trigonometric functions are continuous at every number in their domains.



Derivatives of the Trigonometric Functions

Derivatives of Trigonometric Functions (1 of 9)

If we sketch the graph of the function $f(x) = \sin x$ and use the interpretation of $f'(x)$ as the slope of the tangent to the sine curve in order to sketch the graph of f' , then it looks as if the graph of f' may be the same as the cosine curve. (See Figure 1).

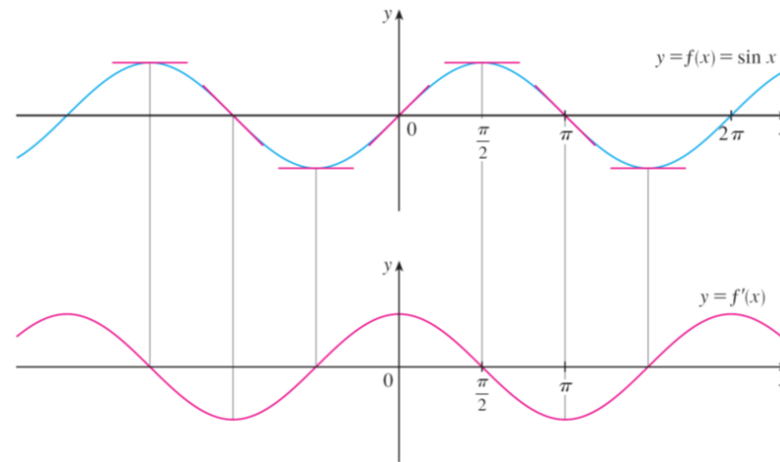


Figure 1

Derivatives of Trigonometric Functions (2 of 9)

Let's try to confirm our guess that if $f(x) = \sin x$, then $f'(x) = \cos x$.

From the definition of a derivative, we have

$$\begin{aligned} f'(x) &= \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} = \lim_{h \rightarrow 0} \frac{\sin(x+h) - \sin x}{h} \\ &= \lim_{h \rightarrow 0} \frac{\sin x \cos h + \cos x \sin h - \sin x}{h} \\ &= \lim_{h \rightarrow 0} \left[\frac{\sin x \cos h - \sin x}{h} + \frac{\cos x \sin h}{h} \right] \end{aligned}$$

Derivatives of Trigonometric Functions (3 of 9)

$$\begin{aligned} &= \lim_{h \rightarrow 0} \left[\sin x \left(\frac{\cos h - 1}{h} \right) + \cos x \left(\frac{\sin h}{h} \right) \right] \\ \textcolor{red}{1} \quad &= \lim_{h \rightarrow 0} \sin x \cdot \lim_{h \rightarrow 0} \frac{\cos h - 1}{h} + \lim_{h \rightarrow 0} \cos x \cdot \lim_{h \rightarrow 0} \frac{\sin h}{h} \end{aligned}$$

Two of these four limits are easy to evaluate. Since we regard x as a constant when computing a limit as $h \rightarrow 0$, we have

$$\lim_{h \rightarrow 0} \sin x = \sin x \quad \text{and} \quad \lim_{h \rightarrow 0} \cos x = \cos x$$

Derivatives of Trigonometric Functions

By the Squeeze Theorem, we have

2

$$\lim_{\theta \rightarrow 0} \frac{\sin \theta}{\theta} = 1$$

Derivatives of Trigonometric Functions

We can deduce the value of the remaining limit in (1) as follows:

$$\begin{aligned}\lim_{\theta \rightarrow 0} \frac{\cos \theta - 1}{\theta} &= \lim_{\theta \rightarrow 0} \left(\frac{\cos \theta - 1}{\theta} \cdot \frac{\cos \theta + 1}{\cos \theta + 1} \right) = \lim_{\theta \rightarrow 0} \frac{\cos^2 \theta - 1}{\theta (\cos \theta + 1)} \\&= \lim_{\theta \rightarrow 0} \frac{-\sin^2 \theta}{\theta (\cos \theta + 1)} = -\lim_{\theta \rightarrow 0} \left(\frac{\sin \theta}{\theta} \cdot \frac{\sin \theta}{\cos \theta + 1} \right) \\&= -\lim_{\theta \rightarrow 0} \frac{\sin \theta}{\theta} \cdot \lim_{\theta \rightarrow 0} \frac{\sin \theta}{\cos \theta + 1} \\&= -1 \cdot \left(\frac{0}{1 + 1} \right) = 0\end{aligned}$$

(by Equation 2)

Derivatives of Trigonometric Functions (4 of 9)

Putting these limits into (1), we get

$$\begin{aligned} f'(x) &= \lim_{h \rightarrow 0} \sin x \cdot \lim_{h \rightarrow 0} \frac{\cos h - 1}{h} + \lim_{h \rightarrow 0} \cos x \cdot \lim_{h \rightarrow 0} \frac{\sin h}{h} \\ &= (\sin x) \cdot 0 + (\cos x) \cdot 1 = \cos x \end{aligned}$$

So we have proved the formula for the derivative of the sine function:

$$2 \quad \frac{d}{dx}(\sin x) = \cos x$$

Example 1

Differentiate $y = x^2 \sin x$.
e

Solution:

Using the Product Rule and Formula 2, we have

$$\begin{aligned}\frac{dy}{dx} &= x^2 \frac{d}{dx}(\sin x) + \sin x \frac{d}{dx}(x^2) \\ &= x^2 \cos x + 2x \sin x\end{aligned}$$

Derivatives of Trigonometric Functions (5 of 9)

Using the same methods as in the proof of Formula 2, one can prove that

$$3 \quad \frac{d}{dx}(\cos x) = -\sin x$$

The tangent function can also be differentiated by using the definition of a derivative, but it is easier to use the Quotient Rule together with Formulas 2 and 3:

$$\frac{d}{dx}(\tan x) = \frac{d}{dx}\left(\frac{\sin x}{\cos x}\right)$$

Derivatives of Trigonometric Functions (6 of 9)

$$\begin{aligned} &= \frac{\cos x \frac{d}{dx}(\sin x) - \sin x \frac{d}{dx}(\cos x)}{\cos^2 x} \\ &= \frac{\cos x \cdot \cos x - \sin x(-\sin x)}{\cos^2 x} \\ &= \frac{\cos^2 x + \sin^2 x}{\cos^2 x} \\ &= \frac{1}{\cos^2 x} = \sec^2 x \end{aligned}$$

Derivatives of Trigonometric Functions (7 of 9)

$$4 \quad \frac{d}{dx}(\tan x) = \sec^2 x$$

The derivatives of the remaining trigonometric functions, csc, sec, and cot, can also be found easily using the Quotient Rule.

Derivatives of Trigonometric Functions (8 of 9)

We collect all the differentiation formulas for trigonometric functions in the following table. Remember that they are valid only when x is measured in radians.

Derivatives of Trigonometric Functions

$$\begin{array}{ll} \frac{d}{dx}(\sin x) = \cos x & \frac{d}{dx}(\csc x) = -\csc x \cot x \\ \frac{d}{dx}(\cos x) = -\sin x & \frac{d}{dx}(\sec x) = \sec x \tan x \\ \frac{d}{dx}(\tan x) = \sec^2 x & \frac{d}{dx}(\cot x) = -\csc^2 x \end{array}$$

Derivatives of Trigonometric Functions (9 of 9)

Trigonometric functions are often used in modeling real-world phenomena. In particular, vibrations, waves, elastic motions, and other quantities that vary in a periodic manner can be described using trigonometric functions. In the next example we discuss an instance of simple harmonic motion.

Example 3

An object fastened to the end of a vertical spring is stretched 4 cm beyond its rest position and released at time $t = 0$. (See Figure 4 and note that the downward direction is positive.) Its position at time t is

$$s = f(t) = 4 \cos t$$

Find the velocity and acceleration at time t and use them to analyze the motion of the object.

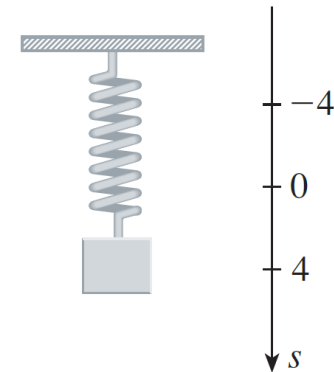


Figure 4

Example 3 – Solution (1 of 3)

The velocity and acceleration are

$$\begin{aligned} v &= \frac{ds}{dt} \\ &= \frac{d}{dt}(4 \cos t) \\ &= 4 \frac{d}{dt}(\cos t) \\ &= -4 \sin t \end{aligned}$$

Example 3 – Solution (2 of 3)

$$\begin{aligned} a &= \frac{dv}{dt} \\ &= \frac{d}{dt}(-4 \sin t) \\ &= -4 \frac{d}{dt}(\sin t) \\ &= -4 \cos t \end{aligned}$$

The object oscillates from the lowest point ($s = 4$ cm) to the highest point ($s = -4$ cm). The period of the oscillation is 2π , the period of $\cos t$.

Example 3 – Solution (3 of 3)

The speed is $|v| = 4 |\sin t|$, which is greatest when $|\sin t| = 1$, that is, when $\cos t = 0$.

So the object moves fastest as it passes through its equilibrium position ($s = 0$). Its speed is 0 when $\sin t = 0$, that is, at the high and low points.

The acceleration $a = -4 \cos t = 0$ when $s = 0$. It has greatest magnitude at the high and low points. See the graphs in Figure 5.

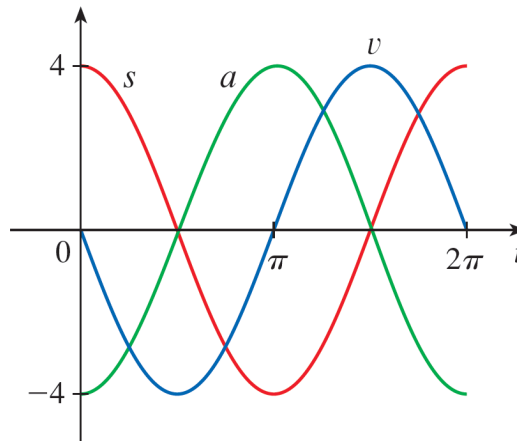


Figure 5

Example 5

Find $\lim_{x \rightarrow 0} \frac{\sin 7x}{4x}$.

变换成熟悉的形式：
 $\sin x/x$

Solution:

In order to apply Equation 5, we first rewrite the function by multiplying and dividing by 7:

$$\frac{\sin 7x}{4x} = \frac{7}{4} \left(\frac{\sin 7x}{7x} \right)$$

If we let $\theta = 7x$, then $\theta \rightarrow 0$ as $x \rightarrow 0$, so by Equation 5 we have

$$\begin{aligned} \lim_{x \rightarrow 0} \frac{\sin 7x}{4x} &= \frac{7}{4} \lim_{x \rightarrow 0} \left(\frac{\sin 7x}{7x} \right) \\ &= \frac{7}{4} \lim_{\theta \rightarrow 0} \frac{\sin \theta}{\theta} = \frac{7}{4} \cdot 1 = \frac{7}{4} \end{aligned}$$



3.4

The Chain Rule

The Chain Rule (1 of 8)

Suppose you are asked to differentiate the function

$$F(x) = \sqrt{x^2 + 1}$$

The differentiation formulas you learned in the previous sections of this chapter enable you to calculate

$$F'(x).$$

Observe that F is a composite function. In fact, if we let $y = f(u) = \sqrt{u}$ and let $u = g(x) = x^2 + 1$, then we can write $y = F(x) = f(g(x))$, that is, $F = f \circ g$.

We know how to differentiate both f and g , so it would be useful to have a rule that tells us how to find the derivative of $F = f \circ g$ in terms of the derivatives of f and g .

The Chain Rule (2 of 8)

It turns out that the derivative of the composite function $f \circ g$ is the product of the derivatives of f and g . This fact is one of the most important of the differentiation rules and is called the *Chain Rule*.

It seems plausible if we interpret derivatives as rates of change. Regard $\frac{du}{dx}$ as the rate of change of u with respect to x , $\frac{dy}{du}$ as the rate of change of y with respect to u , and $\frac{dy}{dx}$ as the rate of change of y with respect to x .

If u changes twice as fast as x and y changes three times as fast as u , then it seems reasonable that y changes six times as fast as x , and so we expect that dy/dx is the product of dy/du and du/dx .

The Chain Rule (3 of 8)

The Chain Rule If g is differentiable at x and f is differentiable at $g(x)$, then the composite function $F = f \circ g$ defined by $F(x) = f(g(x))$ is differentiable at x and F' is given by the product

$$1 \quad F'(x) = f'(g(x)) \cdot g'(x)$$

In Leibniz notation, if $y = f(u)$ and $u = g(x)$ are both differentiable functions, then

$$2 \quad \frac{dy}{dx} = \frac{dy}{du} \frac{du}{dx}$$

The Chain Rule (4 of 8)

Formula 2 is easy to remember because if we think of $\frac{dy}{du}$ and $\frac{du}{dx}$ as quotients, then we could cancel du .

Remember, however, that du has not been defined and $\frac{du}{dx}$ should not be considered of as an actual quotient.

Example 1

Find $F'(x)$ if $F(x) = \sqrt{x^2 + 1}$.

Solution 1:

(Using Formula 1): We have expressed F as $F(x) = (f \circ g) = f(g(x))$ where $f(u) = \sqrt{u}$ and $g(x) = x^2 + 1$.

Since

$$f'(u) = \frac{1}{2} u^{-\frac{1}{2}} = \frac{1}{2\sqrt{u}} \quad \text{and} \quad g'(x) = 2x$$

we have

$$F'(x) = f'(g(x)) \cdot g'(x)$$

$$= \frac{1}{2\sqrt{x^2 + 1}} \cdot 2x = \frac{x}{\sqrt{x^2 + 1}}$$

Example 1 – Solution 2

(Using Formula 2): If we let $u = x^2 + 1$ and $y = \sqrt{u}$, then

$$\begin{aligned} F'(x) &= \frac{dy}{du} \frac{du}{dx} \\ &= \frac{1}{2\sqrt{u}} (2x) \\ &= \frac{1}{2\sqrt{x^2 + 1}} (2x) \\ &= \frac{x}{\sqrt{x^2 + 1}} \end{aligned}$$

The Chain Rule (5 of 8)

When using Formula 2 we should bear in mind that $\frac{dy}{dx}$ refers to the derivative of y when y is considered as a function of x (the derivative of y with respect to x), whereas $\frac{dy}{du}$ refers to the derivative of y when considered as a function of u (the derivative of y with respect to u). For instance, in Example 1, y can be considered as a function of $x = (y = \sqrt{x^2 + 1})$ and also as a function of $u (y = \sqrt{u})$.

Note that

$$\frac{dy}{dx} = F'(x) = \frac{x}{\sqrt{x^2 + 1}} \text{ whereas } \frac{dy}{du} = f'(u) = \frac{1}{2\sqrt{u}}$$

The Chain Rule (6 of 8)

In general, if $y = \sin u$, where u is a differentiable function of x , then, by the Chain Rule,

$$\frac{dy}{dx} = \frac{dy}{du} \frac{du}{dx} = \cos u \frac{du}{dx}$$

Thus

$$\frac{d}{dx}(\sin u) = \cos u \frac{du}{dx}$$

In a similar fashion, all of the formulas for differentiating trigonometric functions can be combined with the Chain Rule.

The Chain Rule (7 of 8)

Let's make explicit the special case of the Chain Rule where the outer function f is a power function.

If $y = [g(x)]^n$, then we can write $y = f(u) = u^n$ where $u = g(x)$. By using the Chain Rule and then the Power Rule, we get

$$\frac{dy}{dx} = \frac{dy}{du} \frac{du}{dx} = n u^{n-1} \frac{du}{dx} = n [g(x)]^{n-1} g'(x)$$

4 The Power Rule Combined with the Chain Rule If n is any real number and $u = g(x)$ is differentiable, then

$$\frac{d}{dx}(u^n) = n u^{n-1} \frac{du}{dx}$$

Alternatively,

$$\frac{d}{dx}[g(x)]^n = n [g(x)]^{n-1} \cdot g'(x)$$

Example 3

Differentiate $y = (x^3 - 1)^{100}$.

Solution:

Taking $u = g(x) = x^3 - 1$ and $n = 100$ in (4), we have

$$\begin{aligned}\frac{dy}{dx} &= \frac{d}{dx} (x^3 - 1)^{100} \\ &= 100 (x^3 - 1)^{99} \frac{d}{dx} (x^3 - 1) \\ &= 100 (x^3 - 1)^{99} \cdot 3x^2 \\ &= 300x^2 (x^3 - 1)^{99}\end{aligned}$$

The Chain Rule (8 of 8)

The reason for the name “Chain Rule” becomes clear when we make a longer chain by adding another link.

Suppose that $y = f(u)$, $u = g(x)$, and $x = h(t)$, where f , g , and h are differentiable functions.

Then, to compute the derivative of y with respect to t , we use the Chain Rule twice:

$$\frac{dy}{dt} = \frac{dy}{dx} \frac{dx}{dt} = \frac{dy}{du} \frac{du}{dx} \frac{dx}{dt}$$



Derivatives of General Exponential Functions

Derivatives of General Exponential Functions (1 of 1)

We can use the Chain Rule to differentiate an exponential function with any base $b > 0$. We know that $a = b^{\ln b}$. So

$$b^x = e^{(\ln b)x}$$

and then the Chain Rule gives

$$\begin{aligned}\frac{d}{dx}(b^x) &= \frac{d}{dx}(e^{(\ln b)x}) = e^{(\ln b)x} \frac{d}{dx}[(\ln b)x] \\ &= e^{(\ln b)x} \ln b = b^x \ln b\end{aligned}$$

because $\ln b$ is a constant. So we have the formula

$$\mathbf{5} \quad \frac{d}{dx}(b^x) = b^x \ln b$$

Example 10

Find the derivative of each of the functions.

$$(a) g(x) = 2^x \quad (b) h(x) = 5x^2$$

Solution:

(a) We use Formula 5 with $b = 2$:

$$g'(x) = \frac{d}{dx}(2^x) = 2^x \ln 2$$

This is consistent with the estimate

$$\frac{d}{dx}(2^x) \approx (0.693)2^x$$

because $\ln 2 \approx 0.693147$.

Example 10 – Solution (1 of 1)

(b) The outer function is an exponential function and the inner function is the squaring function, so we use Formula 5 and the Chain Rule to get

$$h'(x) = \frac{d}{dx}(5^{x^2}) = 5^{x^2} \ln 5 \cdot \frac{d}{dx}(x^2) = 2x \cdot 5^{x^2} \ln 5$$



How to Prove the Chain Rule

How to Prove the Chain Rule (1 of 2)

We know that if $y = f(x)$ and x changes from a to $a + \Delta x$, we defined the increment of y as

$$\Delta y = f(a + \Delta x) - f(a)$$

According to the definition of a derivative, we have

$$\lim_{\Delta x \rightarrow 0} \frac{\Delta y}{\Delta x} = f'(a)$$

So if we denote by ε the difference between $\Delta y / \Delta x$ and $f'(a)$, we obtain

$$\lim_{\Delta x \rightarrow 0} \varepsilon = \lim_{\Delta x \rightarrow 0} \left(\frac{\Delta y}{\Delta x} - f'(a) \right) = f'(a) - f'(a) = 0$$

How to Prove the Chain Rule (2 of 2)

But

$$\varepsilon = \frac{\Delta y}{\Delta x} - f'(a) \quad \Rightarrow \quad \Delta y = f'(a) \Delta x + \varepsilon \Delta x$$

If we define ε to be 0 when $\Delta x = 0$, then ε becomes a continuous function of Δx . Thus, for a differentiable function f , we can write

$$\textcolor{red}{6} \quad \Delta y = f'(a) \Delta x + \varepsilon \Delta x \quad \text{where} \quad \varepsilon \rightarrow 0 \text{ as } \Delta x \rightarrow 0$$

and ε is a continuous function of Δx . This property of differentiable functions is what enables us to prove the Chain Rule.



3.5

Implicit Differentiation



Implicitly Defined Functions

Implicitly Defined Functions (1 of 5)

The functions that we have met so far can be described by expressing one variable explicitly in terms of another variable—for example,

$$y = \sqrt{x^3 + 1} \quad \text{or} \quad y = x \sin x$$

or, in general, $y = f(x)$.

Some functions, however, are defined implicitly by a relation between x and y such as

$$1 \quad x^2 + y^2 = 25$$

or

$$2 \quad x^3 + y^3 = 6xy$$

Implicitly Defined Functions (2 of 5)

In some cases it is possible to solve such an equation for y as an explicit function (or several functions) of x .

For instance, if we solve Equation 1 for y , we get $y = \pm \sqrt{25 - x^2}$, so two of the functions determined by the implicit Equation 1 are
 $f(x) = \sqrt{25 - x^2}$ and $g(x) = -\sqrt{25 - x^2}$.

Implicitly Defined Functions (3 of 5)

The graphs of f and g are the upper and lower semicircles of the circle $x^2 + y^2 = 25$. (See Figure 1.)

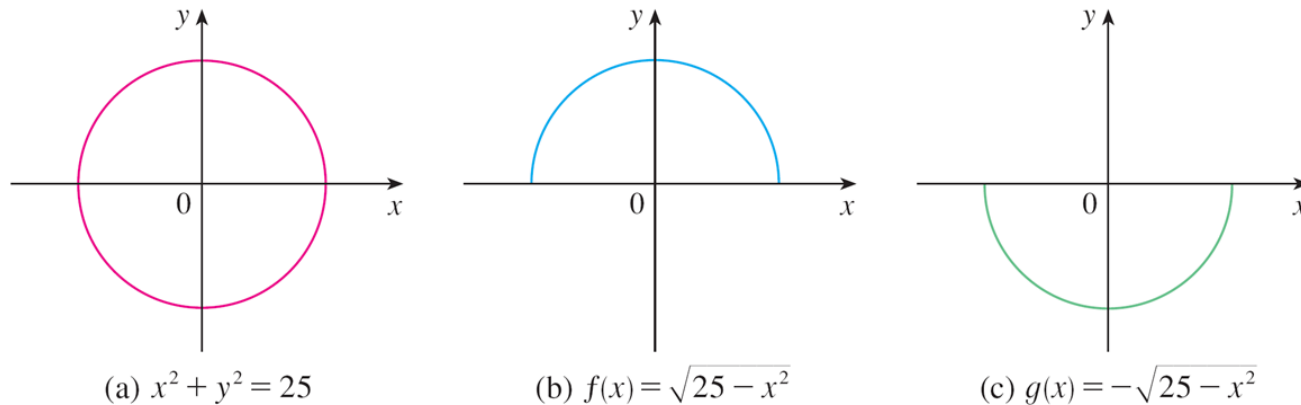
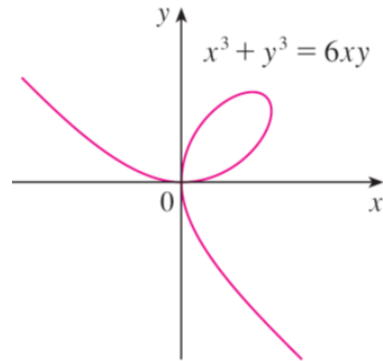


Figure 1

Implicitly Defined Functions (4 of 5)

It's not easy to solve Equation 2 for y explicitly as a function of x by hand. (Even if we use technology, the resulting expressions are very complicated.)

Nonetheless, (2) is the equation of a curve called the **folium of Descartes** shown in Figure 2 and it implicitly defines y as several functions of x .

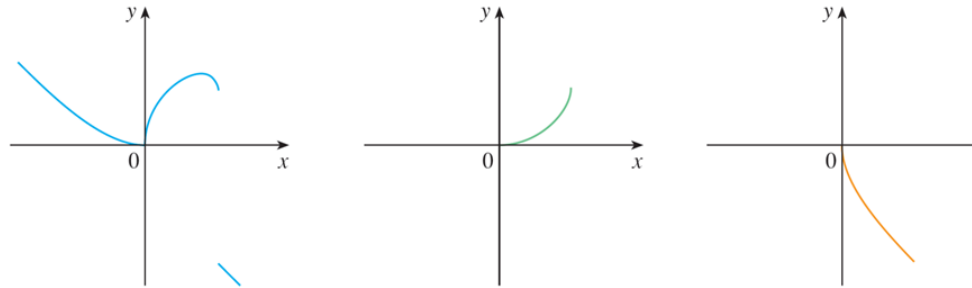


The folium of Descartes

Figure 2

Implicitly Defined Functions (5 of 5)

The graphs of three such functions are shown in Figure 3.



Graphs of three functions defined by the folium of Descartes

Figure 3

When we say that f is a function defined implicitly by Equation 2, we mean that the equation

$$x^3 + [f(x)]^3 = 6xf(x)$$

is true for all values of x in the domain of f .



Implicit Differentiation

Implicit Differentiation (1 of 1)

Fortunately, we don't need to solve an equation for y in terms of x in order to find the derivative of y . Instead we can use the method of **implicit differentiation**.

This consists of differentiating both sides of the equation with respect to x and then solving the resulting equation for dy/dx .

In the examples and exercises of this section it is always assumed that the given equation determines y implicitly as a differentiable function of x so that the method of implicit differentiation can be applied.

Example 1

If $x^2 + y^2 = 25$, find $\frac{dy}{dx}$.

Then find an equation of the tangent to the circle $x^2 + y^2 = 25$ at the point $(3, 4)$.

Solution 1:

Differentiate both sides of the equation $x^2 + y^2 = 25$:

$$\frac{d}{dx}(x^2 + y^2) = \frac{d}{dx}(25)$$

$$\frac{d}{dx}(x^2) + \frac{d}{dx}(y^2) = 0$$

Example 1 – *Solution* (1 of 3)

Remembering that y is a function of x and using the Chain Rule, we have

$$\begin{aligned}\frac{d}{dx}(y^2) &= \frac{d}{dy}(y^2) \frac{dy}{dx} \\ &= 2y \frac{dy}{dx}\end{aligned}$$

Thus

$$2x + 2y \frac{dy}{dx} = 0$$

Now we solve this equation for $\frac{dy}{dx}$:

$$\frac{dy}{dx} = -\frac{x}{y}$$

Example 1 – *Solution* (2 of 3)

At the point (3, 4) we have $x = 3$ and $y = 4$, so

$$\frac{dy}{dx} = -\frac{3}{4}$$

An equation of the tangent to the circle at (3, 4) is therefore

$$y - 4 = -\frac{3}{4}(x - 3) \quad \text{or} \quad 3x + 4y = 25$$

Solution 2:

Solving the equation $x^2 + y^2 = 25$ for y , we get $y = \pm\sqrt{25 - x^2}$.

The point (3, 4) lies on the upper semicircle $y = \sqrt{25 - x^2}$ and so we consider the function $f(x) = \sqrt{25 - x^2}$.

Example 1 – *Solution* (3 of 3)

Differentiating f using the Chain Rule, we have

$$\begin{aligned} f'(x) &= \frac{1}{2}(25 - x^2)^{-\frac{1}{2}} \frac{d}{dx}(25 - x^2) \\ &= \frac{1}{2}(25 - x^2)^{-\frac{1}{2}}(-2x) = -\frac{x}{\sqrt{25 - x^2}} \end{aligned}$$

At the point (3, 4) we have

$$f'(3) = -\frac{3}{\sqrt{25 - 3^2}} = -\frac{3}{4}$$

and, as in Solution 1, an equation of the tangent is $3x + 4y = 25$.



Second Derivatives of Implicit Functions

Second Derivatives of Implicit Functions (1 of 1)

The next example shows how to find the second derivative of a function that is defined implicitly.

Example 4

Find y'' if $x^4 + y^4 = 16$.

Solution:

Differentiating the equation implicitly with respect to x , we get

$$4x^3 + 4y^3 y' = 0$$

Solving for y' gives

$$3 \quad y' = -\frac{x^3}{y^3}$$

Example 4 – Solution (1 of 2)

To find y'' we differentiate this expression for y' using the Quotient Rule and remembering that y is a function of x :

$$\begin{aligned}y'' &= \frac{d}{dx} \left(-\frac{x^3}{y^3} \right) = -\frac{y^3 (d/dx)(x^3) - x^3 (d/dx)(y^3)}{(y^3)^2} \\&= -\frac{y^3 \cdot 3x^2 - x^3 (3y^2 y')}{y^6}\end{aligned}$$

If we now substitute Equation 3 into this expression, we get

$$y'' = -\frac{3x^2 y^3 - 3x^3 y^2 \left(-\frac{x^3}{y^3} \right)}{y^6}$$

Example 4 – Solution (2 of 2)

$$= -\frac{3(x^2 y^4 + x^6)}{y^7} = \frac{3x^2(y^4 + x^4)}{y^7}$$

But the values of x and y must satisfy the original equation $x^4 + y^4 = 16$.

So the answer simplifies to

$$y'' = -\frac{3x^2(16)}{y^7} = -48 \frac{x^2}{y^7}$$



3.6

Derivatives of Logarithmic and Inverse Trigonometric Functions



Derivatives of Logarithmic Functions

Derivatives of Logarithmic Functions (1 of 2)

Because the logarithmic function $y = \log_b x$ is the inverse of the exponential function $y = b^x$, which we know is differentiable from Section 3.1, it follows that the logarithmic function is also differentiable.

We now state and prove the formula for the derivative of a logarithmic function.

$$y = \log_b x$$

$$x = b^y$$

Derivatives of Logarithmic Functions (2 of 2)

$$1 \quad \frac{d}{dx}(\log_b x) = \frac{1}{x \ln b}$$

$$2 \quad \frac{d}{dx}(\ln x) = \frac{1}{x}$$

In general, if we combine Formula 2 with the Chain Rule, we get

$$3 \quad \frac{d}{dx}(\ln u) = \frac{1}{u} \frac{du}{dx} \quad \text{or} \quad \frac{d}{dx}[\ln g(x)] = \frac{g'(x)}{g(x)}$$

Example 2

Find

$$\frac{d}{dx} \ln (\sin x).$$

Solution:

Using (3), we have

$$\begin{aligned} \frac{d}{dx} \ln (\sin x) &= \frac{1}{\sin x} \frac{d}{dx} (\sin x) &= \frac{1}{\sin x} \cos x \\ & &= \cot x \end{aligned}$$



Logarithmic Differentiation

Logarithmic Differentiation (1 of 2)

The calculation of derivatives of complicated functions involving products, quotients, or powers can often be simplified by taking logarithms.

The method used in the next example is called **logarithmic differentiation**.

Example 7

Differentiate

$$y = \frac{x^{\frac{3}{4}} \sqrt{x^2 + 1}}{(3x + 2)^5}.$$

Solution:

We take logarithms of both sides of the equation and use the Laws of Logarithms to simplify:

$$\ln y = \frac{3}{4} \ln x + \frac{1}{2} \ln (x^2 + 1) - 5 \ln (3x + 2)$$

Differentiating implicitly with respect to x gives

$$\frac{1}{y} \frac{dy}{dx} = \frac{3}{4} \cdot \frac{1}{x} + \frac{1}{2} \cdot \frac{2x}{x^2 + 1} - 5 \cdot \frac{3}{3x + 2}$$

Example 7 – *Solution*

Solving for $\frac{dy}{dx}$, we get

$$\frac{dy}{dx} = y \left(\frac{3}{4x} + \frac{x}{x^2 + 1} - \frac{15}{3x + 2} \right)$$

Because we have an explicit expression for y , we can substitute and write

$$\frac{dy}{dx} = \frac{x^{\frac{3}{4}} \sqrt{x^2 + 1}}{(3x + 2)^5} \left(\frac{3}{4x} + \frac{x}{x^2 + 1} - \frac{15}{3x + 2} \right)$$

Logarithmic Differentiation (2 of 2)

Steps in Logarithmic Differentiation

1. Take natural logarithms of both sides of an equation $y = f(x)$ and use the Laws of Logarithms to expand the expression.
2. Differentiate implicitly with respect to x .
3. Solve the resulting equation for y' and replace y by $f(x)$.

We know that the general version of the Power Rule states that if n is any real number and $f(x) = x^n$, then $f'(x) = nx^{n-1}$.



The Number e as a Limit

The Number e as a Limit (1 of 5)

If $f(x) = \ln x$, then $f'(x) = \frac{1}{x}$. Thus $f'(1) = 1$. We now use this fact to express the number e as a limit.

From the definition of a derivative as a limit, we have

$$\begin{aligned} f'(1) &= \lim_{h \rightarrow 0} \frac{f(1+h) - f(1)}{h} = \lim_{x \rightarrow 0} \frac{f(1+x) - f(1)}{x} \\ &= \lim_{x \rightarrow 0} \frac{\ln(1+x) - \ln 1}{x} = \lim_{x \rightarrow 0} \frac{1}{x} \ln(1+x) \\ &= \lim_{x \rightarrow 0} \ln(1+x)^{\frac{1}{x}} \end{aligned}$$

The Number e as a Limit (2 of 5)

Because $f'(1) = 1$, we have

$$\lim_{x \rightarrow 0} \ln(1 + x)^{\frac{1}{x}} = 1$$

Then, by Theorem 2.5.8,

If f is continuous at b and $\lim_{x \rightarrow a} g(x) = b$, then $\lim_{x \rightarrow a} f(g(x)) = f(b)$.

In other words,

$$\lim_{x \rightarrow a} f(g(x)) = f\left(\lim_{x \rightarrow a} g(x)\right)$$

The Number e as a Limit (3 of 5)

And the continuity of the exponential function, we have

$$\begin{aligned} e = e^1 &= e^{\lim_{x \rightarrow 0} \ln(1+x) \frac{1}{x}} &= \lim_{x \rightarrow 0} e^{\ln(1+x) \frac{1}{x}} \\ & &= \lim_{x \rightarrow 0} (1+x)^{\frac{1}{x}} \\ \mathbf{5} \quad e &= \lim_{x \rightarrow 0} (1+x)^{\frac{1}{x}} \end{aligned}$$

The Number e as a Limit (4 of 5)

Formula 5 is illustrated by the graph of the function $y = (1 + x)^{\frac{1}{x}}$ in Figure 4 and a table of values for small values of x . This illustrates the fact that, correct to seven decimal places,

$$e \approx 2.7182818$$

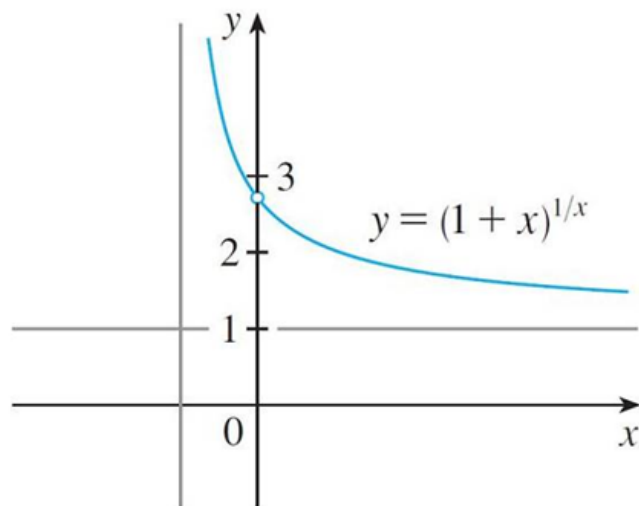


Figure 4

x	$(1 + x)^{1/x}$
0.1	2.59374246
0.01	2.70481383
0.001	2.71692393
0.0001	2.71814593
0.00001	2.71826824
0.000001	2.71828047
0.0000001	2.71828169
0.00000001	2.71828181

The Number e as a Limit (5 of 5)

If we put $n = \frac{1}{x}$ in Formula 5, then $n \rightarrow \infty$ as $x \rightarrow 0^+$ and so an alternative expression for e is

$$6 \quad e = \lim_{n \rightarrow \infty} \left(1 + \frac{1}{n} \right)^n$$



Derivatives of Inverse Trigonometric Functions

Derivatives of Inverse Trigonometric Functions (1 of 4)

We know that the definition of the arcsine function:

$$y = \sin^{-1} x \quad \text{means} \quad \sin y = x \quad \text{and} \quad -\frac{\pi}{2} \leq y \leq \frac{\pi}{2}$$

Differentiating $\sin y = x$ implicitly with respect to x , we obtain

$$\cos y \frac{dy}{dx} = 1 \quad \text{or} \quad \frac{dy}{dx} = \frac{1}{\cos y}$$

Now $\cos y \geq 0$, since $-\frac{\pi}{2} \leq y \leq \frac{\pi}{2}$, so

$$\cos y = \sqrt{1 - \sin^2 y} = \sqrt{1 - x^2} \quad (\cos^2 y + \sin^2 y = 1)$$

Derivatives of Inverse Trigonometric Functions (2 of 4)

Therefore

$$\frac{dy}{dx} = \frac{1}{\cos y} = \frac{1}{\sqrt{1-x^2}}$$
$$\frac{d}{dx}(\sin^{-1} x) = \frac{1}{\sqrt{1-x^2}}$$

The formula for the derivative of the arctangent function is derived in a similar way. If $y = \tan^{-1} x$, then $\tan y = x$. Differentiating $\tan y = x$ implicitly with respect to x , we have

$$\sec^2 y \frac{dy}{dx} = 1$$

Derivatives of Inverse Trigonometric Functions (3 of 4)

$$\frac{dy}{dx} = \frac{1}{\sec^2 y} = \frac{1}{1 + \tan^2 y} = \frac{1}{1 + x^2}$$

$$\frac{d}{dx}(\tan^{-1} x) = \frac{1}{1 + x^2}$$

Derivatives of Inverse Trigonometric Functions (4 of 4)

The derivatives of the remaining four are given in the following table.

Derivatives of Inverse Trigonometric Functions

$$\begin{array}{ll}\frac{d}{dx}(\sin^{-1} x) = \frac{1}{\sqrt{1-x^2}} & \frac{d}{dx}(\csc^{-1} x) = -\frac{1}{x\sqrt{x^2-1}} \\ \frac{d}{dx}(\cos^{-1} x) = -\frac{1}{\sqrt{1-x^2}} & \frac{d}{dx}(\sec^{-1} x) = \frac{1}{x\sqrt{x^2-1}} \\ \frac{d}{dx}(\tan^{-1} x) = \frac{1}{1+x^2} & \frac{d}{dx}(\cot^{-1} x) = -\frac{1}{1+x^2}\end{array}$$

Example 9

Differentiate

(a) $y = \frac{1}{\sin^{-1} x}$ and

(b) $f(x) = x \arctan \sqrt{x}$.

Solution:

$$\begin{aligned} \text{(a)} \quad \frac{dy}{dx} &= \frac{d}{dx} (\sin^{-1} x)^{-1} \\ &= -(\sin^{-1} x)^{-2} \frac{d}{dx} (\sin^{-1} x) \\ &= -\frac{1}{(\sin^{-1} x)^2 \sqrt{1-x^2}} \end{aligned}$$

Example 9 – Solution

$$\begin{aligned} \text{(b) } f'(x) &= x \frac{1}{1 + (\sqrt{x})^2} \left(\frac{1}{2} x^{-\frac{1}{2}} \right) + \arctan \sqrt{x} \\ &= \frac{\sqrt{x}}{2(1+x)} + \arctan \sqrt{x} \end{aligned}$$



3.7

Rates of Change in the Natural and Social Sciences

Rates of Change in the Natural and Social Sciences (1 of 3)

We know that if $y = f(x)$, then the derivative $\frac{dy}{dx}$ can be interpreted as the rate of change of y with respect to x . In this section we examine some of the applications of this idea to physics, chemistry, biology, economics, and other sciences.

We know that the basic idea behind rates of change. If x changes from x_1 to x_2 , then the change in x is

$$\Delta x = x_2 - x_1$$

and the corresponding change in y is

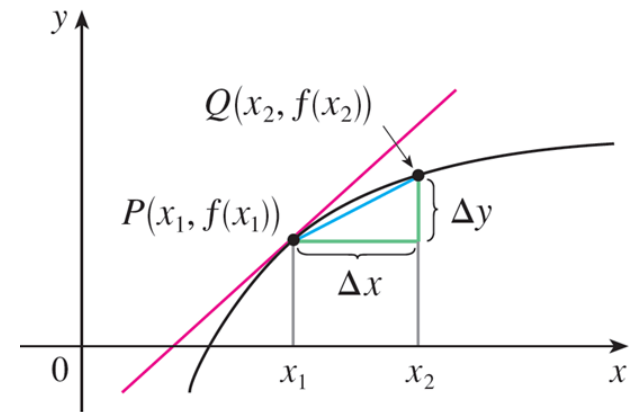
$$\Delta y = f(x_2) - f(x_1)$$

Rates of Change in the Natural and Social Sciences (2 of 3)

The difference quotient

$$\frac{\Delta y}{\Delta x} = \frac{f(x_2) - f(x_1)}{x_2 - x_1}$$

is the **average rate of change of y with respect to x** over the interval $[x_1, x_2]$ and can be interpreted as the slope of the secant line PQ in Figure 1.



m_{PQ} = average rate of change

$m = f'(x_1)$ = instantaneous rate of change

Figure 1

Rates of Change in the Natural and Social Sciences (3 of 3)

Its limit as $\Delta x \rightarrow 0$ is the $f'(x_1)$, which can therefore be interpreted as the instantaneous rate of change of y with respect to x or the slope of the tangent line at $P(x_1, f(x_1))$.

Using Leibniz notation, we write the process in the form

$$\frac{dy}{dx} = \lim_{\Delta x \rightarrow 0} \frac{\Delta y}{\Delta x}$$



Physics

Physics (1 of 1)

If $s = f(t)$ is the position function of a particle that is moving in a straight line, then $\frac{\Delta s}{\Delta t}$ represents the average velocity over a time period Δt , and $v = \frac{ds}{dt}$ represents the instantaneous **velocity** (the rate of change of displacement with respect to time). The instantaneous rate of change of velocity with respect to time is **acceleration**: $a(t) = v'(t) = s''(t)$.

Now that we know the differentiation formulas, we are able to more easily solve problems involving the motion of objects.

Example 1 (1 of 2)

The position of a particle is given by the equation

$$s = f(t) = t^3 - 6t^2 + 9t$$

where t is measured in seconds and s in meters.

- (a) Find the velocity at time t .
- (b) What is the velocity after 2 s? After 4 s?
- (c) When is the particle at rest?
- (d) When is the particle moving forward (that is, in the positive direction)?

Example 1 (2 of 2)

- (e) Draw a diagram to represent the motion of the particle.
- (f) Find the total distance traveled by the particle during the first five seconds.
- (g) Find the acceleration at time t and after 4 s.
- (h) Graph the position, velocity, and acceleration functions for $0 \leq t \leq 5$.
- (i) When is the particle speeding up? When is it slowing down?

Example 1 – Solution (1 of 10)

(a) The velocity function is the derivative of the position function:

$$s = f(t) = t^3 - 6t^2 + 9t$$

$$v(t) = \frac{ds}{dt} = 3t^2 - 12t + 9$$

Example 1 – Solution (2 of 10)

(b) The velocity after 2 s means the instantaneous velocity when $t = 2$, that is,

$$\begin{aligned}v(2) &= \left. \frac{ds}{dt} \right|_{t=2} = 3(2)^2 - 12(2) + 9 \\&= -3 \text{ m/s}\end{aligned}$$

The velocity after 4 s is

$$\begin{aligned}v(4) &= 3(4)^2 - 12(4) + 9 \\&= 9 \text{ m/s}\end{aligned}$$

Example 1 – Solution (3 of 10)

(c) The particle is at rest when $v(t) = 0$, that is,

$$\begin{aligned}3t^2 - 12t + 9 &= 3(t^2 - 4t + 3) \\&= 3(t - 1)(t - 3) \\&= 0\end{aligned}$$

and this is true when $t = 1$ or $t = 3$.

Thus the particle is at rest at 1 s and 3 s, respectively.

Example 1 – Solution (4 of 10)

(d) The particle moves in the positive direction when $v(t) > 0$, that is,

$$3t^2 - 12t + 9 = 3(t - 1)(t - 3) > 0$$

This inequality is true when both factors are positive ($t > 3$) or when both factors are negative ($t < 1$).

Thus the particle moves in the positive direction in the time intervals $t < 1$ and $t > 3$.

It moves backward (in the negative direction) when $1 < t < 3$.

Example 1 – Solution (5 of 10)

(e) Using the information from part (d) we make a schematic sketch in Figure 2 of the motion of the particle back and forth along a line (the s -axis).

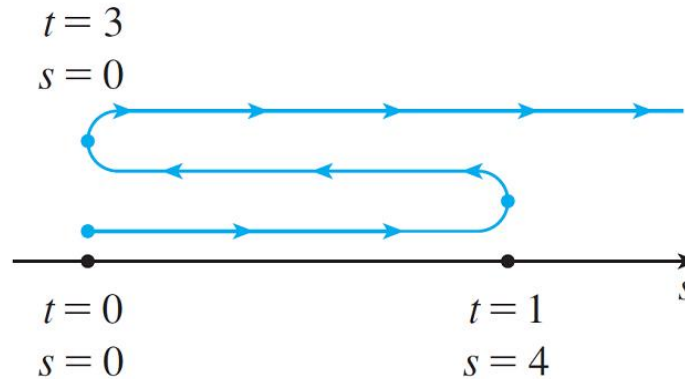


Figure 2

Example 1 – Solution (6 of 10)

(f) Because of what we learned in parts (d) and (e), we need to calculate the distances traveled during the time intervals $[0, 1]$, $[1, 3]$, and $[3, 5]$ separately.

The distance traveled in the first second is $|f(1) - f(0)| = |4 - 0| = 4 \text{ m}$

From $t = 1$ to $t = 3$ the distance traveled is $|f(3) - f(1)| = |0 - 4| = 4 \text{ m}$

From $t = 3$ to $t = 5$ the distance traveled is $|f(5) - f(3)| = |20 - 0| = 20 \text{ m}$

The total distance is $4 + 4 + 20 = 28 \text{ m}$.

Example 1 – Solution (7 of 10)

(g) The acceleration is the derivative of the velocity function:

$$\begin{aligned}a(t) &= \frac{d^2 s}{dt^2} \\&= \frac{dv}{dt} \\&= 6t - 12\end{aligned}$$

$$\begin{aligned}a(4) &= 6(4) - 12 \\&= 12 \text{ m/s}^2\end{aligned}$$

Example 1 – Solution (8 of 10)

(h) Figure 3 shows the graphs of s , v , and a .

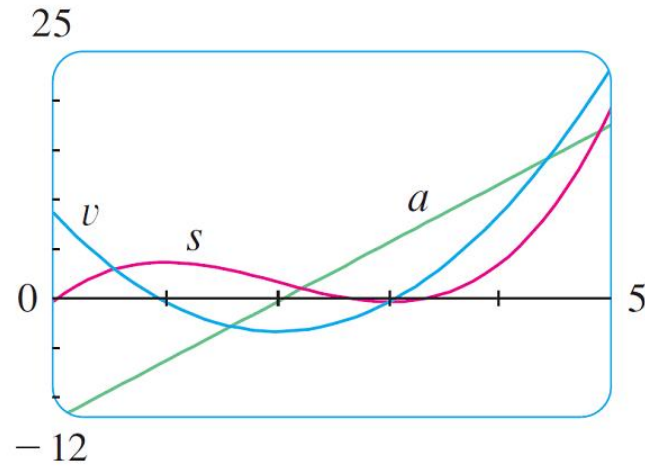


Figure 3

Example 1 – Solution (9 of 10)

- (i) The particle speeds up when the velocity is positive and increasing (v and a are both positive) and also when the velocity is negative and decreasing (v and a are both negative).

In other words, the particle speeds up when the velocity and acceleration have the same sign. (The particle is pushed in the same direction it is moving.)

From Figure 3 we see that this happens when $1 < t < 2$ and when $t > 3$.

Example 1 – Solution (10 of 10)

The particle slows down when v and a have opposite signs, that is, when $0 \leq t < 1$ and when $2 < t < 3$.

Figure 4 summarizes the motion of the particle.

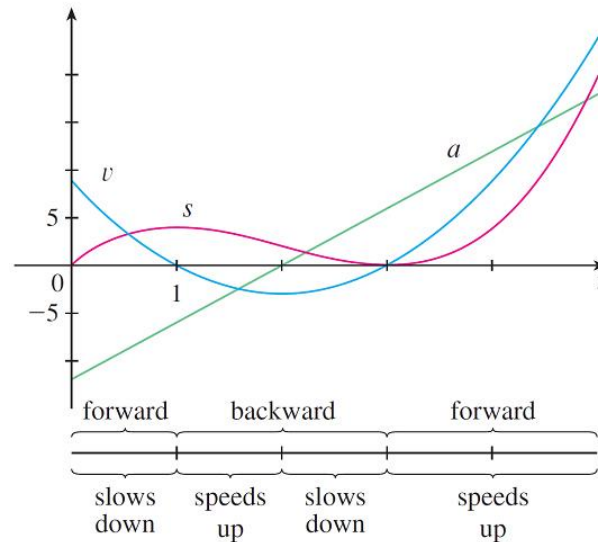


Figure 4

Example 2 (1 of 4)

If a rod or piece of wire is homogeneous, then its *linear density* is uniform and is defined as the mass per unit length $\left(\rho = \frac{m}{l}\right)$ and measured in kilograms per meter.

Suppose, however, that the rod is not homogeneous but that its mass measured from its left end to a point x is $m = f(x)$, as shown in Figure 5.

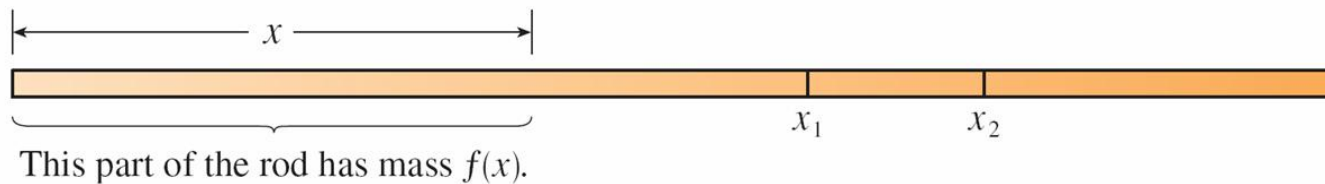


Figure 5

Example 2 (2 of 4)

The mass of the part of the rod that lies between $x = x_1$ and $x = x_2$ is given by $\Delta m = f(x_2) - f(x_1)$, so the average density of that part of the rod is

$$\text{average density} = \frac{\Delta m}{\Delta x} = \frac{f(x_2) - f(x_1)}{x_2 - x_1}$$

If we now let $\Delta x \rightarrow 0$ (that is, $x_2 \rightarrow x_1$), we are computing the average density over smaller and smaller intervals.

The **linear density** ρ at x_1 is the limit of these average densities as $\Delta x \rightarrow 0$; that is, the linear density is the rate of change of mass with respect to length.

Example 2 (3 of 4)

Symbolically,

$$\rho = \lim_{\Delta x \rightarrow 0} \frac{\Delta m}{\Delta x} = \frac{dm}{dx}$$

Thus the linear density of the rod is the derivative of mass with respect to length.

For instance, if $m = f(x) = \sqrt{x}$, where x is measured in meters and m in kilograms, then the average density of the part of the rod given by $1 \leq x \leq 1.2$ is

$$\frac{\Delta m}{\Delta x} = \frac{f(1.2) - f(1)}{1.2 - 1} = \frac{\sqrt{1.2} - 1}{0.2}$$

Example 2 (4 of 4)

$$\approx 0.48 \text{ kg/m}$$

while the density right at $x = 1$ is

$$\begin{aligned}\rho &= \left. \frac{dm}{dx} \right|_{x=1} = \left. \frac{1}{2\sqrt{x}} \right|_{x=1} \\ &= 0.50 \text{ kg/m}\end{aligned}$$

Example 3 (1 of 2)

A current exists whenever electric charges move. Figure 6 shows part of a wire and electrons moving through a plane surface, shaded red.

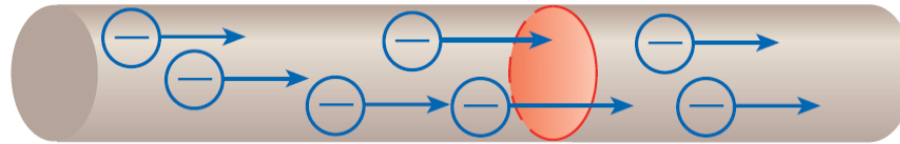


Figure 6

If ΔQ is the net charge that passes through this surface during a time period Δt , then the average current during this time interval is defined as

$$\text{average current} = \frac{\Delta Q}{\Delta t} = \frac{Q_2 - Q_1}{t_2 - t_1}$$

Example 3 (2 of 2)

If we take the limit of this average current over smaller and smaller time intervals, we get what is called the **current** I at a given time t_1 :

$$I = \lim_{\Delta t \rightarrow 0} \frac{\Delta Q}{\Delta t} = \frac{dQ}{dt}$$

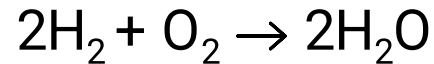
Thus the current is the rate at which charge flows through a surface. It is measured in units of charge per unit time (often coulombs per second, called amperes).



Chemistry

Example 4 (1 of 5)

A chemical reaction results in the formation of one or more substances (called *products*) from one or more starting materials (called *reactants*). For instance, the “equation”



indicates that two molecules of hydrogen and one molecule of oxygen form two molecules of water.

Let’s consider the reaction



where A and B are the reactants and C is the product.

Example 4 (2 of 5)

The **concentration** of a reactant A is the number of moles (1 mole = 6.022×10^{23} molecules) per liter and is denoted by [A].

The concentration varies during a reaction, so [A], [B], and [C] are all functions of time (t).

The average rate of reaction of the product C over a time interval $t_1 \leq t \leq t_2$ is

$$\frac{\Delta [C]}{\Delta t} = \frac{[C](t_2) - [C](t_1)}{t_2 - t_1}$$

Example 4 (3 of 5)

But chemists are interested in the **instantaneous rate of reaction** because it gives information about the mechanism of the chemical reaction. The instantaneous rate of reaction is obtained by taking the limit of the average rate of reaction as the time interval Δt approaches 0:

$$\text{rate of reaction} = \lim_{\Delta t \rightarrow 0} \frac{\Delta [C]}{\Delta t} = \frac{d[C]}{dt}$$

Since the concentration of the product increases as the reaction proceeds, the derivative $\frac{d[C]}{dt}$ will be positive, and so the rate of reaction of C is positive.

Example 4 (4 of 5)

The concentrations of the reactants, however, decrease during the reaction, so, to make the rates of reaction of A and B positive numbers, we put minus signs

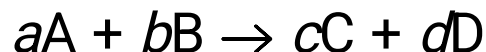
in front of the derivatives $\frac{d[A]}{dt}$ and $\frac{d[B]}{dt}$.

Since [A] and [B] each decrease at the same rate that [C] increases, we have

$$\text{rate of reaction} = \frac{d[C]}{dt} = -\frac{d[A]}{dt} = -\frac{d[B]}{dt}$$

Example 4 (5 of 5)

More generally, it turns out that for a reaction of the form



we have

$$-\frac{1}{a} \frac{d[A]}{dt} = -\frac{1}{b} \frac{d[B]}{dt} = \frac{1}{c} \frac{d[C]}{dt} = \frac{1}{d} \frac{d[D]}{dt}$$

The rate of reaction can be determined from data and graphical methods. In some cases there are explicit formulas for the concentrations as functions of time that enable us to compute the rate of reaction.

Example 5 (1 of 4)

One of the quantities of interest in thermodynamics is compressibility. If a given substance is kept at a constant temperature, then its volume V depends on its pressure P .

We can consider the rate of change of volume with respect to pressure—namely, the derivative $\frac{dV}{dP}$. As P increases, V decreases, so $\frac{dV}{dP} < 0$.

The **compressibility** is defined by introducing a minus sign and dividing this derivative by the volume V :

$$\text{isothermal compressibility} = \beta = -\frac{1}{V} \frac{dV}{dP}$$

Example 5 (2 of 4)

Thus β measures how fast, per unit volume, the volume of a substance decreases as the pressure on it increases at constant temperature.

For instance, the volume V (in cubic meters) of a sample of air at 25°C was found to be related to the pressure P (in kilopascals) by the equation

$$V = \frac{5.3}{P}$$

Example 5 (3 of 4)

The rate of change of V with respect to P when $P = 50$ kPa is

$$\begin{aligned}\left. \frac{dV}{dP} \right|_{P=50} &= - \left. \frac{5.3}{P^2} \right|_{P=50} \\ &= - \frac{5.3}{2500} \\ &= -0.00212 \text{ m}^3 / \text{kPa}\end{aligned}$$

Example 5 (4 of 4)

The compressibility at that pressure is

$$\begin{aligned}\beta &= - \frac{1}{V} \frac{dV}{dP} \bigg|_{P=50} \\ &= \frac{0.00212}{\frac{5.3}{50}} \\ &= 0.02 \text{ (m}^3 \text{ / kPa) / m}^3\end{aligned}$$



Biology

Example 6 (1 of 6)

Let $n = f(t)$ be the number of individuals in an animal or plant population at time t .

The change in the population size between the times $t = t_1$ and $t = t_2$ is Δn
 $= f(t_2) - f(t_1)$, and so the average rate of growth during the time period $t_1 \leq t \leq t_2$ is

$$\text{average rate of growth} = \frac{\Delta n}{\Delta t} = \frac{f(t_2) - f(t_1)}{t_2 - t_1}$$

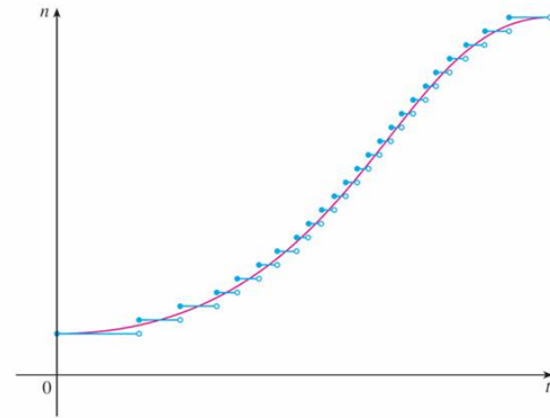
The **instantaneous rate of growth** is obtained from this average rate of growth by letting the time period Δt approach 0:

$$\text{growth rate} = \lim_{\Delta t \rightarrow 0} \frac{\Delta n}{\Delta t} = \frac{dn}{dt}$$

Example 6 (2 of 6)

Strictly speaking, this is not quite accurate because the actual graph of a population function $n = f(t)$ would be a step function that is discontinuous whenever a birth or death occurs and therefore not differentiable.

However, for a large animal or plant population, we can replace the graph by a smooth approximating curve as in Figure 7.



A smooth curve approximating a growth function

Figure 7

Example 6 (3 of 6)

To be more specific, consider a population of bacteria in a homogeneous nutrient medium.

Suppose that by sampling the population at certain intervals it is determined that the population doubles every hour.

If the initial population is n_0 and the time t is measured in hours, then

$$f(1) = 2f(0) = 2n_0$$

$$f(2) = 2f(1) = 2^2 n_0$$

Example 6 (4 of 6)

$$f(3) = 2f(2) = 2^3 n_0$$

and, in general,

$$f(t) = 2^t n_0$$

The population function
is

$$n = n_0 2^t.$$

We have shown that

$$\frac{d}{dx}(b^x) = b^x \ln b$$

Example 6 (5 of 6)

So the rate of growth of the bacteria population at time t is

$$\begin{aligned}\frac{dn}{dt} &= \frac{d}{dt}(n_0 2^t) \\ &= n_0 2^t \ln 2\end{aligned}$$

For example, suppose that we start with an initial population of $n_0 = 100$ bacteria. Then the rate of growth after 4 hours is

$$\left. \frac{dn}{dt} \right|_{t=4} = 100 \cdot 2^4 \ln 2$$

Example 6 (6 of 6)

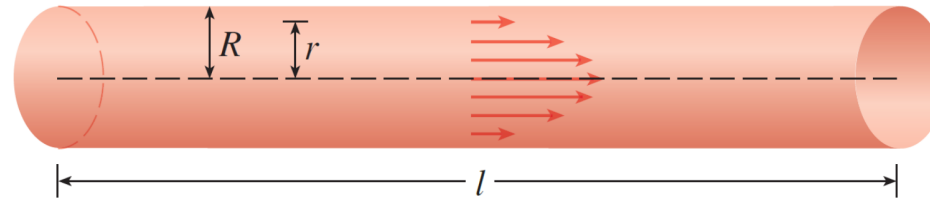
$$= 1600 \ln 2$$

$$\approx 1109$$

This means that, after 4 hours, the bacteria population is growing at a rate of about 1109 bacteria per hour.

Example 7 (1 of 6)

When we consider the flow of blood through a blood vessel, such as a vein or artery, we can model the shape of the blood vessel by a cylindrical tube with radius R and length l as illustrated in Figure 8.



Blood flow in an artery

Figure 8

Because of friction at the walls of the tube, the velocity v of the blood is greatest along the central axis of the tube and decreases as the distance r from the axis increases until v becomes 0 at the wall.

Example 7 (2 of 6)

The relationship between v and r is given by the **law of laminar flow**, which was experimentally derived by the French physicist Jean Léonard Marie Poiseuille in 1838.

This law states that

$$1 \quad v = \frac{P}{4\eta l} (R^2 - r^2)$$

where η is the viscosity of the blood and P is the pressure difference between the ends of the tube.

If P and l are constant, then v is a function of r with domain $[0, R]$.

Example 7 (3 of 6)

The average rate of change of the velocity as we move from $r = r_1$ outward to $r = r_2$ is given by

$$\frac{\Delta v}{\Delta r} = \frac{v(r_2) - v(r_1)}{r_2 - r_1}$$

and if we let $\Delta r \rightarrow 0$, we obtain the **velocity gradient**, that is, the instantaneous rate of change of velocity with respect to r :

$$\text{velocity gradient} = \lim_{\Delta r \rightarrow 0} \frac{\Delta v}{\Delta r} = \frac{dv}{dr}$$

Example 7 (4 of 6)

Using Equation 1, we obtain

$$\begin{aligned}\frac{dv}{dr} &= \frac{P}{4\eta l}(0 - 2r) \\ &= -\frac{Pr}{2\eta l}\end{aligned}$$

For one of the smaller human arteries we can take $\eta = 0.027$, $R = 0.008$ cm, $l = 2$ cm, and $P = 4000$ dynes/cm², which gives

$$v = \frac{4000}{4(0.027)^2}(0.000064 - r^2)$$

Example 7 (5 of 6)

$$\approx 1.85 \times 10^4 (6.4 \times 10^{-5} - r^2)$$

At $r = 0.02$ cm the blood is flowing at a speed of

$$\begin{aligned} v(0.002) &\approx 1.85 \times 10^4 (6.4 \times 10^{-6} - 4 \times 10^{-6}) \\ &= 1.11 \text{ cm/s} \end{aligned}$$

and the velocity gradient at that point is

$$\left. \frac{dv}{dr} \right|_{r=0.002} = - \frac{4000(0.002)}{2(0.027)^2} \approx -74 \text{ (cm/s) / cm}$$

Example 7 (6 of 6)

To get a feeling for what this statement means, let's change our units from centimeters to micrometers ($1 \text{ cm} = 10,000 \text{ }\mu\text{m}$). Then the radius of the artery is $80 \text{ }\mu\text{m}$.

The velocity at the central axis is $11,850 \text{ }\mu\text{m/s}$, which decreases to $11,110 \text{ }\mu\text{m/s}$ at a distance of $r = 20 \text{ }\mu\text{m}$.

The fact that $\frac{dv}{dr} = -74 (\text{ }\mu\text{m/s}) / \text{ }\mu\text{m}$ means that, when $r = 20 \text{ }\mu\text{m}$, the velocity is decreasing at a rate of about $74 \text{ }\mu\text{m/s}$ for each micrometer that we proceed away from the center.



Economics

Example 8 (1 of 6)

Suppose $C(x)$ is the total cost that a company incurs in producing x units of a certain commodity.

The function C is called a **cost function**. If the number of items produced is increased from x_1 to x_2 , then the additional cost is $\Delta C = C(x_2) - C(x_1)$, and the average rate of change of the cost is

$$\begin{aligned}\frac{\Delta C}{\Delta x} &= \frac{C(x_2) - C(x_1)}{x_2 - x_1} \\ &= \frac{C(x_1 + \Delta x) - C(x_1)}{\Delta x}\end{aligned}$$

Example 8 (2 of 6)

The limit of this quantity as $\Delta x \rightarrow 0$, that is, the instantaneous rate of change of cost with respect to the number of items produced, is called the **marginal cost** by economists:

$$\text{m a r g i n a l c o s t} = \lim_{\Delta x \rightarrow 0} \frac{\Delta C}{\Delta x} = \frac{dC}{dx}$$

[Since x often takes on only integer values, it may not make literal sense to let Δx approach 0, but we can always replace $C(x)$ by a smooth approximating function as in Example 6.]

Example 8 (3 of 6)

Taking $\Delta x = 1$ and n large (so that Δx is small compared to n), we have

$$C'(n) \approx C(n+1) - C(n)$$

Thus the marginal cost of producing n units is approximately equal to the cost of producing one more unit [the $(n+1)$ st unit].

Example 8 (4 of 6)

It is often appropriate to represent a total cost function by a polynomial

$$C(x) = a + bx + cx^2 + dx^3$$

where a represents the overhead cost (rent, heat, maintenance) and the other terms represent the cost of raw materials, labor, and so on. (The cost of raw materials may be proportional to x , but labor costs might depend partly on higher powers of x because of overtime costs and inefficiencies involved in large-scale operations.)

Example 8 (5 of 6)

For instance, suppose a company has estimated that the cost (in dollars) of producing x items is

$$C(x) = 10,000 + 5x + 0.01x^2$$

Then the marginal cost function is

$$C'(x) = 5 + 0.02x$$

The marginal cost at the production level of 500 items is

$$\begin{aligned} C'(500) &= 5 + 0.02(500) \\ &= \$15/\text{item} \end{aligned}$$

Example 8 (6 of 6)

This gives the rate at which costs are increasing with respect to the production level when $x = 500$ and predicts the cost of the 501st item.

The actual cost of producing the 501st item is

$$\begin{aligned} C(501) - C(500) &= \left[10,000 + 5(501) + 0.01(501)^2 \right] - \left[10,000 + 5(500) + 0.01(500)^2 \right] \\ &= \$15.01 \end{aligned}$$

Notice that $C'(500) \approx C(501) - C(500)$.



Other Sciences

Other Sciences (1 of 1)

Rates of change occur in all the sciences. A geologist is interested in knowing the rate at which an intruded body of molten rock cools by conduction of heat into surrounding rocks.

An engineer wants to know the rate at which water flows out of a reservoir.

An urban geographer is interested in the rate of change of the population density in a city as the distance from the city center increases.

A meteorologist is concerned with the rate of change of atmospheric pressure with respect to height.



A Single Idea, Many Interpretations

A Single Idea, Many Interpretations (1 of 2)

Velocity, density, current, power, and temperature gradient in physics; rate of reaction and compressibility in chemistry; rate of growth and blood velocity gradient in biology; marginal cost and marginal profit in economics; rate of heat flow in geology; rate of improvement of performance in psychology; rate of spread of a rumor in sociology—these are all special cases of a single mathematical concept, the derivative.

All of these different applications of the derivative illustrate the fact that part of the power of mathematics lies in its abstractness.

A Single Idea, Many Interpretations (2 of 2)

A single abstract mathematical concept (such as the derivative) can have different interpretations in each of the sciences.

When we develop the properties of the mathematical concept once and for all, we can then turn around and apply these results to all of the sciences.

This is much more efficient than developing properties of special concepts in each separate science.